



EchoSync

AI-Assisted Home Safety for Seniors Living Alone

Privacy-first pre-arrival intelligence for faster community response

1 PROBLEM STATEMENT

Unwitnessed home emergencies can go undetected when seniors live alone



No Bystander

A collapse may go unnoticed when the senior is alone at home



Manual Alerts May Fail

Panic buttons can be unreachable, forgotten or unused during emergencies



Delayed Detection Increases Risk

Late alerts can delay caregiver, SCDF or community response



2 SOLUTION SUMMARY

Echosync combines ambient sensing, edge AI, voice check-ins, caregiver support, and SCDF-led response



Ambient sensing

Motion/Presence cues without cameras; sound, motion, and non-camera cues



Voice check-in

Asks : "EchoSync Check in, Are you okay?"



Community Help + SCDF escalation

Serious cases are reviewed by SCDF, who can activate myResponder/ nearby helpers when needed



Caregiver verification

Trusted next-of-kin review; low/medium alerts can be verified; high/critical alerts continue to operator review



Privacy-first: No raw video. No facial recognition. Only encrypted alert summaries + node status shared.

3 ARCHITECTURE / WORKFLOW

1 Sensing



Sound, motion, presence, thermal cues

2 Edge AI



Dell Pro Max Micro GB10

Llama 3.1 8B Instruct

3 Voice Check-In



"Hello Mdm/Sir, are you okay?"

4 Risk Routing



Alerts are classified and routed based on severity.

- Low - Minimal concern. Log only
- Medium - Potential issue. Caregiver App alert.
- High - Urgent. Escalated to SCDF review.
- Critical - Immediate danger. SCDF dispatch.

5 SCDF Response



SCDF reviews high-risk alerts and activates myResponder / 995 if needed.



End-to-end encrypted



No CCTV



No raw audio stored



GCC-Ready

4 UNIQUE FEATURES



Multi-Signal AI Fusion

Combines thermal, motion, sound, and presence cues.



Confidence Scoring

Reduces false alerts through smarter filtering before escalation.



Privacy-First Design

No CCTV, no raw audio, only alert summaries and node status shared.



Caregiver + SCDF Coordination

Clear separation of roles for faster, safer response.

5



EchoSync



Built for privacy, speed, and practical community response.

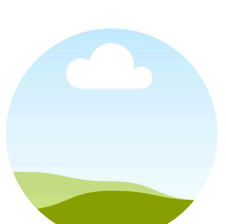


Smarter alerts. Faster help. Stronger community



SCAN TO VIEW ECHOSYNC PROTOTYPE

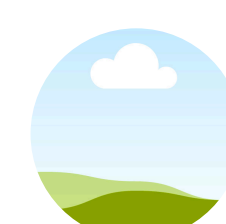
TEAM MEMBERS



Vivion Oh



Felicia Tan



Charlisa Tan